

Multiple Choice (10 marks)

1. Which of the following is a biotic factor that could affect the growth rate of a population?
 - (a) Flash flood
 - (b) Tsunami
 - (c) Sudden reduction in the animal food resource
 - (d) Destruction of the ozone layer
 - (e) Drought
2. Which of the following statements about trypsin is NOT true?
 - (a) It is an organic compound made of proteins.
 - (b) It is a catalyst that alters the rate of a reaction.
 - (c) It is operative over a wide pH range.
 - (d) The rate of catalysis is affected by the concentration of substrate.
3. Certain nerve gases are known to cause a breakdown of cholinesterase. How would these gases affect the human body and why?
 - (a) Cause euphoria and hallucinations
 - (b) Improve memory and cognitive functions
 - (c) Increase physical strength and endurance
 - (d) Cause tremors, spasms, or even death
 - (e) Trigger temporary paralysis without pain
4. Why does milk "when kept in a refrigerator?
 - (a) Enzymatic reactions with oxygen
 - (b) Lactose crystallization
 - (c) Loss of refrigeration power over time
 - (d) mesophiles
 - (e) Natural separation of milk components without bacterial influence
5. All of the following are examples of events that can prevent interspecific breeding EXCEPT
 - (a) the potential mates experience geographic isolation
 - (b) the potential mates experience behavioral isolation
 - (c) the potential mates have different courtship rituals
 - (d) the potential mates have similar breeding seasons

True / False (10 marks)

Answer True or False

6. True or False: The probability of drawing an ace first and a face card second, without replacement, is $4 / 221$.
7. True or False: Polytene chromosomes in *Drosophila silvarentis* consist of multiple copies of a single chromosome, visible under a light microscope.
8. True or False: Animal cells contain centrioles within centrosomes, which are involved in cell division.
9. True or False: Deserts have high rates of leaching and nutrient cycling due to their limited vegetation and rainfall.
10. True or False: Sexual dimorphism is primarily caused by artificial selection, leading to significant trait differences between sexes.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the expected phenotypic ratios of offspring from the mating of a black male cat and a tortoiseshell female, considering the sex-linked inheritance of coat color genes.
12. Discuss the key characteristics of arthropods that have contributed to their evolutionary success, analyzing their roles in adaptation and survival.

End of Paper